

# Guangxu Yang (杨广绪)

PhD Student in Computer Science

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Location: Los Angeles, USA

## Education

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**University of Southern California (USC)**

*Ph.D. Student in Computer Science*

Advisor: Prof. Jiapeng Zhang

Los Angeles, USA

Aug 2022 – Current

**University of Electronic Science and Technology of China (UESTC)**

*M.E. in Information and Communication Engineering*

*B.E. in Network Engineering*

Chengdu, China

Sep 2019 – Jun 2022

Sep 2015 – Jun 2019

## Research Interests

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- Communication complexity
- Quantum computing

## Publications

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*Note: Authors in theoretical computer science are listed alphabetically.*

### Conference Proceedings

- [1] **Exponential Separation of Quantum and Classical One-Way Numbers-on-Forehead Communication**  
Guangxu Yang, Jiapeng Zhang.  
*67th Annual Symposium on Foundations of Computer Science (FOCS 2026)*.
- [2] **A Lifting Theorem for Hybrid Classical-Quantum Communication Complexity**  
Xudong Wu, Guangxu Yang, Penghui Yao.  
*53rd EATCS International Colloquium on Automata, Languages, and Programming (ICALP 2026)*.
- [3] **A Min-Entropy Approach to Multi-Party Communication Lower Bounds**  
Mi-Ying Huang, Xinyu Mao, Shuo Wang, Guangxu Yang, Jiapeng Zhang.  
*40th Computational Complexity Conference (CCC 2025)*.
- [4] **Gadgetless Lifting Beats Round Elimination: Improved Lower Bounds for Pointer Chasing**  
Xinyu Mao, Guangxu Yang, Jiapeng Zhang.  
*16th Innovations in Theoretical Computer Science (ITCS 2025)*.
- [5] **Communication Lower Bounds for Collision Problems via Density Increment Arguments**  
Guangxu Yang and Jiapeng Zhang.  
*56th Annual ACM Symposium on Theory of Computing (STOC 2024)*.
- [6] **Communication Lower Bounds of Key-Agreement Protocols via Density Increment Arguments**  
Mi-Ying Huang, Xinyu Mao, Guangxu Yang, Jiapeng Zhang.  
*21st Theory of Cryptography Conference (TCC 2023)*.

### Preprints & Manuscripts

- [7] **Deterministic Lifting Theorems for One-Way Number-on-Forehead Communication**  
Guangxu Yang, Jiapeng Zhang. [arXiv:2506.12420](https://arxiv.org/abs/2506.12420).

- [8] **Exponential Quantum Space Advantage for Approximating Max- $k$ -SAT in the Streaming Setting**  
Haoyu Wang, **Guangxu Yang**. [arXiv:2606.05366](#)
- [9] **Prediction Under Imperfect Compression: A Theory of Approximate MDL**  
Qian Li, Xinyu Mao, Shanghua Teng, **Guangxu Yang**. [arXiv:2606.04834](#)
- [10] **Quantum Communication Lower Bounds for Search Problems via Matrix Discrepancy**  
Minbo Gao, Chenghua Liu, **Guangxu Yang**, Tianyi Zhang. [arXiv:2607.08517](#)

## Teaching Experience

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### Teaching Assistant at USC

- **CSCI 570:** Analysis of Algorithms *Spring 2026*
- **CSCI 170:** Discrete Methods in Computer Science *Spring 2025*
- **CSCI 575:** Quantum Computing and Quantum Cryptography *Spring 2024*

## Professional Service

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**Conference Reviewer:** SODA 2027, MFCS 2026, FOCS 2026, PODS 2026, CRYPTO 2025, ITCS 2025, COCOON 2024.

## Presentations

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### Conference Talks

#### Communication Lower Bounds of Key-Agreement Protocols

TCC 2023, Taipei Nov 2023

#### Gadgetless Lifting Beats Round Elimination

ITCS 2025, New York Jan 2025

### Invited Seminar & Group Talks

#### Communication Lower Bounds for Collision Problems via Density Increment Arguments

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|------------------------------------------------|----------|
| University of Southern California Theory Lunch | Oct 2023 |
| Nanjing University                             | Dec 2023 |
| Shanghai Jiao Tong University                  | May 2024 |
| Shanghai University of Finance and Economics   | Oct 2025 |

#### Deterministic Lifting Theorems for One-Way Number-on-Forehead Communication

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|--------------------------------------------------------|----------|
| University of Southern California Theory Lunch         | May 2025 |
| University of Science and Technology of China          | Nov 2025 |
| SoCal Theory Day at University of California San Diego | Feb 2026 |
| Shanghai Jiao Tong University                          | Mar 2026 |